
CS246: Database Management Systems Lab

Week # 08 (1 Questions, 87 Marks)

Lab session: AL1

Held on: 03-Mar-2025 (Mon)

Lab Timings: 14:00 to 17:00 Hours Pages: 4

Submission time: 16:45 Hrs, 03-Mar-2025

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1. Some (sub) tasks are designed to emit errors when SQL statements are invoked. You should make an informed effort to find the cause of the errors. You do not need to resolve the errors.
2. This lab assignment is based on the concepts
 - (a) Section 4.2 Views
 - (b) Section 4.4.6 assigning names to constraints
 - (c) Section 4.5 SQL data types
 - (d) Section 4.5.2 Type conversions and formatting functions

from the text book titled Database System Concepts by Abraham Silberschatz, Henry F Korth & S. Sudarshan

3. Refer to view statement
4. Refer to updatable view statements
5. Refer to the view with check points

Question 1: (87 points)

Using MySQL perform the following tasks:

Task 01 (1 mark) Create a database named *week08* and create following tables in this database.

Task 02 (6 marks) **Create the following tables**

1. (2 marks) A table **sailors** containing the following

column #	name	data type	constraint
1	sid	int	primary key
2	sname	char(50)	None
3	rating	int	None
4	age	decimal(3,1)	None

2. (2 marks) A table **boats** containing the following

column #	name	data type	constraint
1	bid	int	primary key
2	bname	char(50)	None
3	bcolor	char(50)	None

3. (2 marks) A table **reserves** containing the following

column #	name	data type	constraint
1	sid	int	refers to sailors
2	bid	int	refers to boats
3	day	char(50)	None
sid, bid, day is primary key			

Task 03 (3 marks) Populate data

- populate **sailors** table from the file **sailors.csv**
- populate **boats** table from the file **boats.csv**
- populate **reserves** table from the file **reserves.csv**

Task 04 (77 marks) Creating Views

(19 marks) Creating updatable views

- (6 marks) Perform the following
 - (3 marks) Create a view containing the columns **sid**, **rating** from **sailors**.
 - (1 mark) insert the following records into this view with the following values (91, 7), (92, 8), (93, 9), (94, 10), (22, 8)
 - (1 mark) update the rating of **sid=91** to 8.
 - (1 mark) delete the sailor with **sid=91**
- (6 marks) Perform the following
 - (3 marks) Create a view containing the following: **boats** where color is green
 - (1 mark) insert the following records into this view with the following values (205, River Mania, green), (206, green-bird, green), (207, blue-warriors, blue)
 - (1 mark) Create a new view which restricts the insertion into the view with mis-matched color by adding the option **with check option**
 - (1 mark) insert the following records into this view with the following values (207, blue-warriors, blue)
- (7 marks) Perform the following
 - (3 marks) create a view having the columns **sid**, **rating**, **bid**, **bname**
 - (1 mark) insert values (80, 8) for **sid**, **rating** into this view
 - (1 mark) insert values (105, 'Lucky Lake') for **bid**, **bname** into this view
 - (1 mark) update **bname** to Jumper for **bid = 101** in this view
 - (1 mark) update **bname** to Interlake for **bid = 101** in this view

(31 marks) Creating views that are not updatable

- (7 marks) Perform the following

- (3 marks) **view involving aggregate functions** create a view with the columns `sid`, `sname`, `rating`, `bid`, `bname` such that sailors having maximum `rating` among the reserved boats.
 - (1 mark) Insert a record in this view with fields `sid`, `sname`, `rating` taking values `sid = 80`, `sname='best sailor'`, `rating = 10`
 - (1 mark) Update the `rating` of `sid = 74` to `-9` in this view.
 - (1 mark) Delete the record from this view with `sid` having value `74`.
 - (1 mark) Update the `bname` of `bid = 102` to `Can I get updated?` in this view.
2. (6 marks) Perform the following
- (3 marks) **view involving distinct clause** create a view of sailors having distinct `ratings` and obtain the column `rating`.
 - (1 mark) Insert a new `rating` whose value is `2` in the created view
 - (1 mark) Update the `rating` from `7` to `-7` from the created view
 - (1 mark) Delete the `rating 7` from the created view
3. (6 marks) Perform the following
- (3 marks) **view involving group by** create a view for the query: Boats reserved by sailors with same `rating`. Obtain all columns from the tables which are involved in the `join` operation.
 - (1 mark) Insert a new record in the created view having values `sid = 80`, `sname='budding sailor'`, `rating=10`, `age=25`
 - (1 mark) Update the `rating` from `8` to `6` in the created view
 - (1 mark) Delete the `rating 7` from the created view
4. (6 marks) Perform the following
- (3 marks) **view involving having clause** create a view for the query: Boats reserved by sailors with same `rating` whose `age` is more than `36` years. Obtain all columns from the tables which are involved in the `join` operation.
 - (1 mark) Insert a new record in the created view having values `sid = 80`, `sname='budding sailor'`, `rating=10`, `age=25`
 - (1 mark) Update the `rating` from `8` to `6` in the created view
 - (1 mark) Delete the `rating 7` from the created view
5. (6 marks) Perform the following
- (3 marks) **view involving union** create a view which contains sailors who reserved green boat or sailors who reserved blue boat. Include the following columns in the view `sid`, `sname`, `bid`, `bcolor`.
 - (1 mark) Insert a new record in the created view having values `sid = 81`, `sname='union view insert'`, `bid=110`, `bcolor='green'`
 - (1 mark) Update the `sname` of `sid = 81` to `"union view update"`.
 - (1 mark) Delete the `sid=22` from the created view.

(9 marks) **Creating views using views**

1. (3 marks) Create a view to obtain `rating` from the view in Task 04 (1).
2. (1 mark) insert the following records into this view with the following values (7), (8), (9), (10), (8)
3. (1 mark) update the `rating` 8 to 9.
4. (1 mark) delete the `rating` 10.
5. (3 marks) Create a view to obtain `sid`, `bname`, `day` using `reserves` and view created in Task 04 (3).

(9 marks) **Effect of altering original tables on views**

1. (1 mark) Rename the column `rating` of `sailors` table to `rting`
2. (2 marks) List the contents all the above views that involve the column name `rating`
3. (1 mark) Rename the column `rting` of `sailors` to `rating`
4. (2 marks) List the contents all the above views that involve the column name `rating`
5. (1 mark) Delete the column `rating` of `sailors` table.
6. (2 marks) List the contents all the above views that involve the column name `rating`

(9 marks) **Type conversion**

1. (3 marks) **Create tables** `sailors_1`, `reserves_1` and `boats_1` as described in Task 02.
2. (3 marks) Populate data from `sailors.csv` into table `sailors_1`; `reserves.csv` into `reserves_1` table and `boats.csv` into `boats_1` table.
3. (3 marks) Change the following columns data type
 - (a) `sid` to `smallint`
 - (b) `bid` to `char(3)`
 - (c) `bcolor` to `char(5)`